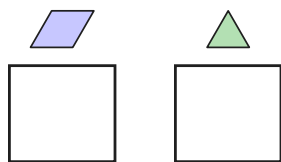
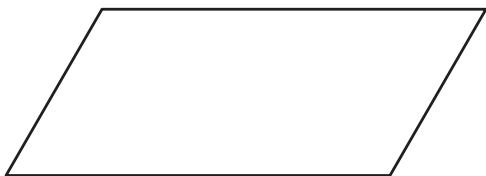
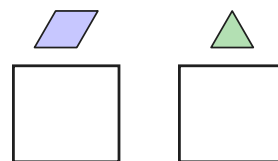
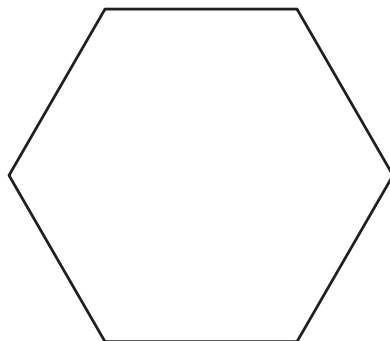
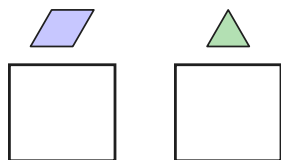
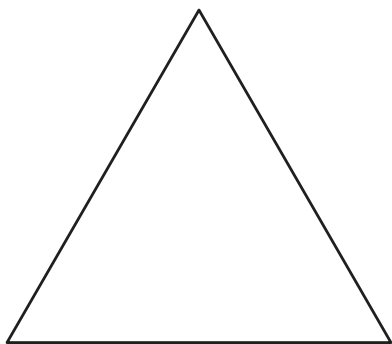
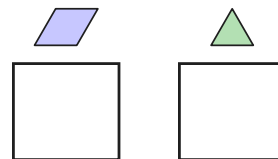
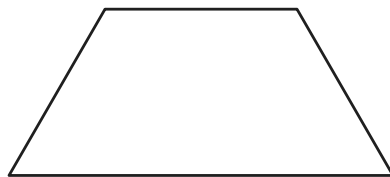
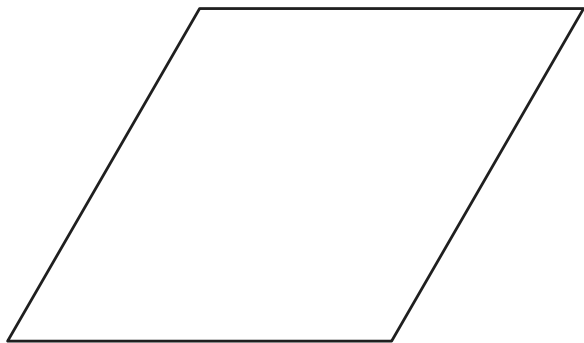


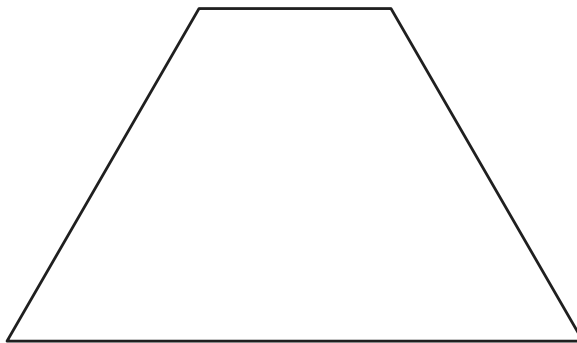
Problem 1: Cover each board with blue rhombi  only. Pieces must not overlap, leave gaps, or stick out past the outline. If you can cover the board, write $\checkmark$ in its first box; if you cannot, write $\times$. Then cover the same board with green triangles  and write how many you used in its second box.

A**B****C****D**

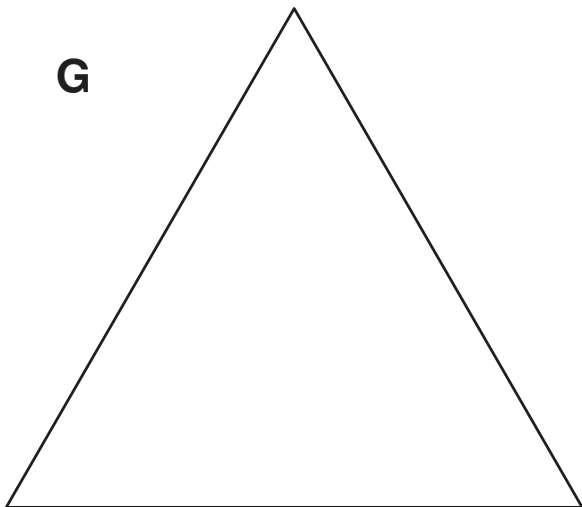
Problem 2: Do the same for boards E, F, G and H: blue rhombi  only, then $\checkmark$ or $\times$ in the first box; then green triangles , and the number in the second box.

E

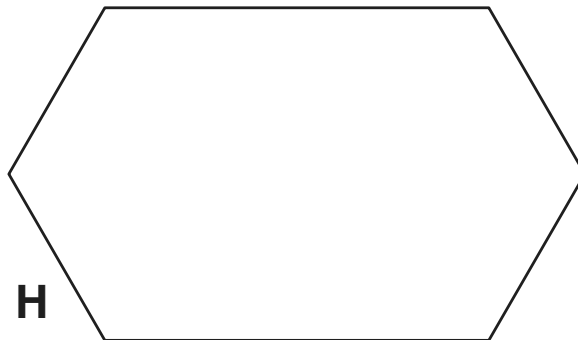
	
☐	☐

F

	
☐	☐

G

	
☐	☐

H

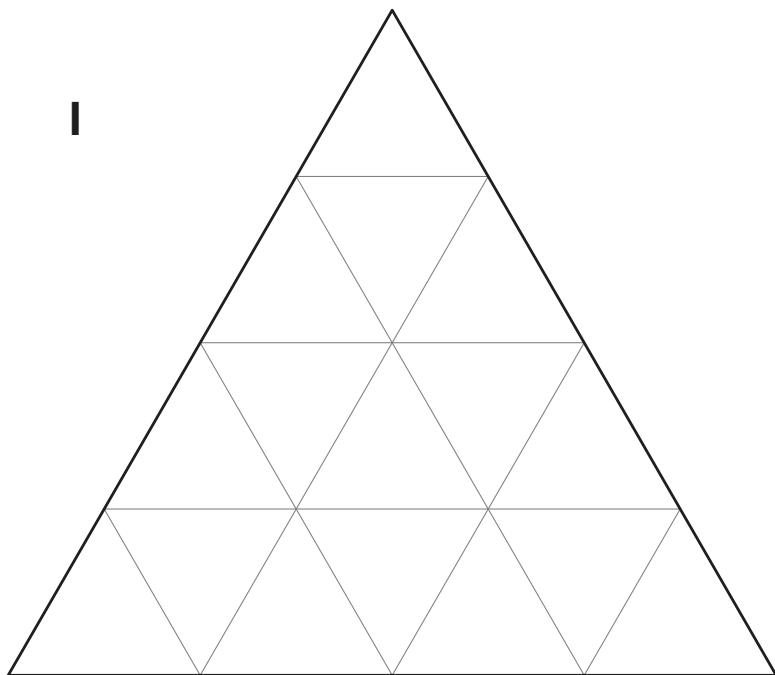
	
☐	☐

Problem 3: Copy your results into the table: for each board, the number of green triangles and $\checkmark$ or $\times$. Circle every board whose number is odd. Then find two boards with the same number where one is $\checkmark$ and the other is $\times$, and write their letters on the line.

board	green triangles	$\checkmark$ or $\times$
A		
B		
C		
D		
E		
F		
G		
H		

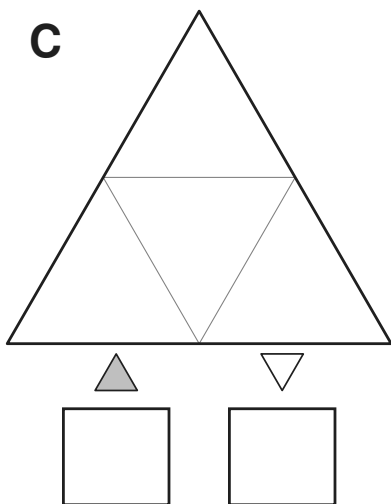
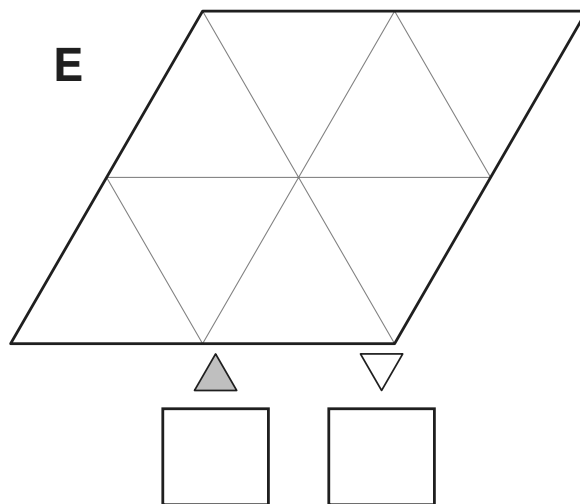
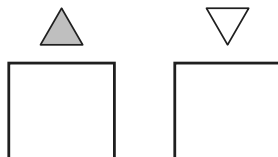
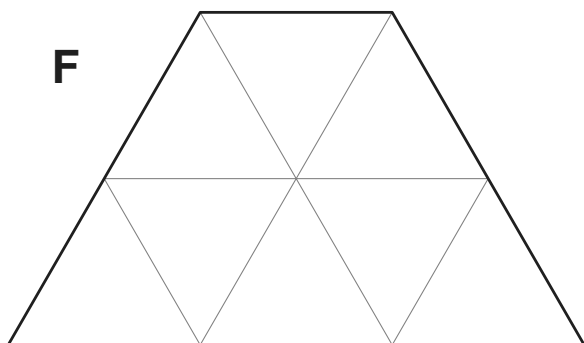
Problem 4: Board I is the big triangle below. Color every up-pointing triangle $\triangle$ of board I and leave every down-pointing triangle ∇ white. A blue rhombus  can lie on board I in three directions. Put it on the board in each direction, and each time write in the table how many colored and how many white triangles it covers.

I



piece		
 direction 1		
 direction 2		
 direction 3		

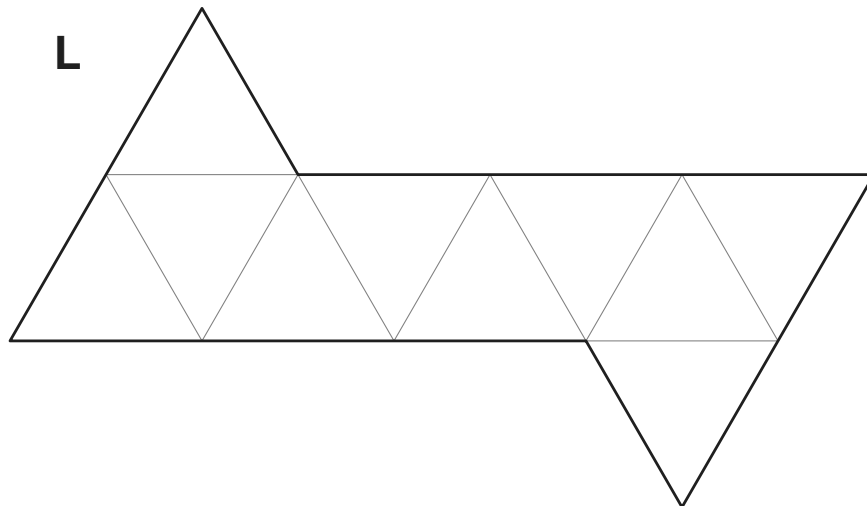
Problem 5: Color the up-pointing triangles $\triangle$ in boards C, E and F. Count the colored and the white triangles in each board and write the two numbers in its boxes. Then explain on the lines why board F cannot be covered by blue rhombi.

C**E****F**

Problem 6: Board I is on page 3; you colored it in Problem 4. Count its colored triangles and its white triangles and write the two numbers in the table. Then, before touching any pieces, write your prediction in the table: ✓ if you think blue rhombi can cover board I, ✕ if you think they cannot. Then check with blue rhombi on board I and write what happened in the last column.

board			prediction	result
I				

Problem 7: Board L is below. Color its up-pointing triangles , count the colored and the white triangles, and write the two numbers in the table. Write your prediction, ✓ or ✕, before touching any pieces. Then check with blue rhombi and write what happened. If your prediction was wrong, write on the lines what you found when you tried to cover board L.

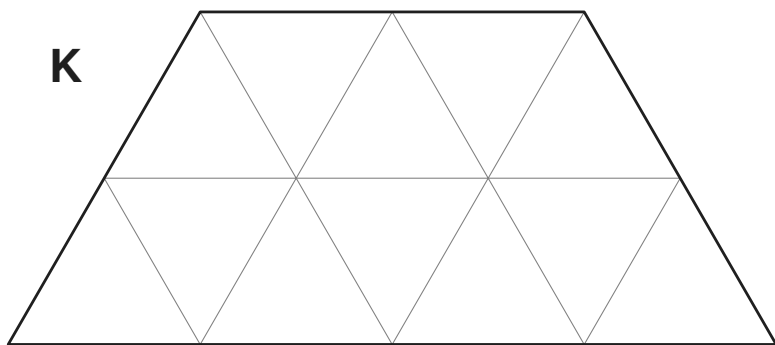


board			prediction	result
L				

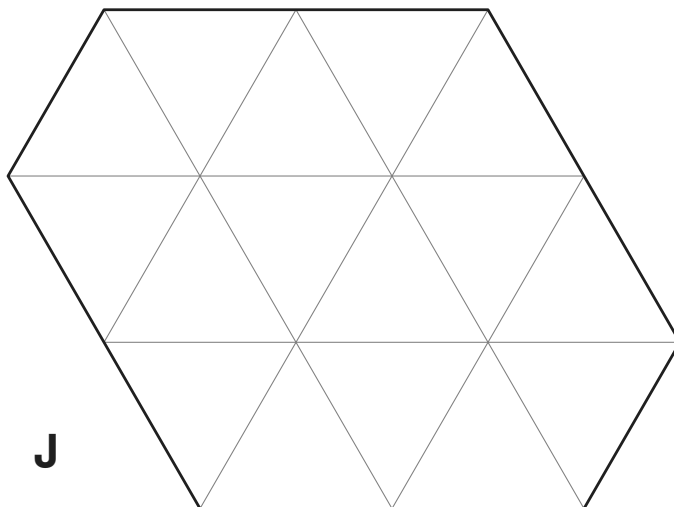
Problem 8: Use board I on page 3 again. Put a red trapezoid  on it with its long side at the bottom, and write in the table how many colored and how many white triangles it covers. Then put it with its long side at the top and write the numbers. Do the same with a green triangle  pointing up and then pointing down, and with a yellow hexagon. Then look at this table and the table in Problem 4: which pieces cover the same number of colored and white triangles every time? Write their colors on the line.

piece		
 long side down		
 long side up		
 pointing up		
 pointing down		
 hexagon		

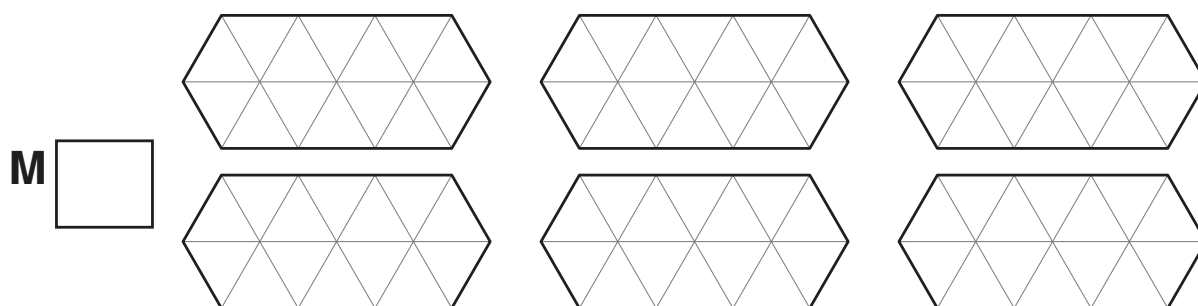
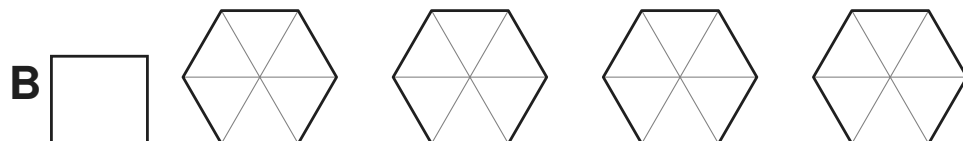
Problem 9: Do the same as in Problems 6 and 7 for boards K and J: color the up-pointing triangles , count, write your prediction before touching any pieces, then check with blue rhombi and write what happened.

K

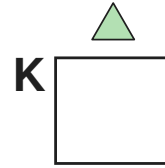
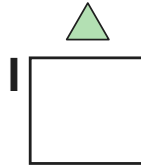
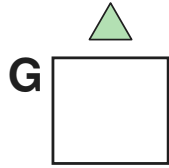
board			prediction	result
K				
J				

J

Problem 10: Board B can be covered by three blue rhombi in more than one way. Find every different way. Two ways count as different if some rhombus is in a different place, even if one is the other turned around. Draw each way on one of the small copies of B by outlining each rhombus with a dark line, and write the number of ways in the box. Then do the same for board H (five rhombi) and for board M (seven rhombi). Board M is printed below at full size. There are more small copies than you need.



Problem 11: Cover board G (page 2) using blue rhombi  and green triangles  together, with as few green triangles as you can. Write your smallest number in the box for G. Do the same for board I (page 3) and board K (page 6). Then explain on the lines why nobody can cover board G with fewer green triangles than your number.



Problem 12: Which of the boards A to K can be covered with red trapezoids  only? Try each board and write ✓ or × in the box under its letter. Then write on the line one thing that is true about the number of green triangles of every board you could cover.

